Iason **Skylitsis**

+30 6943582400 | iason.sky@gmail.com | iasonsky | iason-sky

Work Experience

Twiki Solutions — *Rotterdam, Netherlands*

Junior Software Engineer — *Jun. 2023 - Present*

- Key role in migrating OpenRPA scrapers to Python (with BeautifulSoup and Selenium), achieving a 20% accuracy boost and 7x speed improvement in Twiki Container Tracker v3.
- Leading the Data Driven Services project, developing two Plotly Dash dashboards to monitor Twiki Power container tracker performance (data completeness, accuracy) and analyze carrier performance (lead time, transshipment frequency), providing actionable insights for clients.

Amsterdam UMC — *Amsterdam, Netherlands*

AI Research Intern [paper] — *Oct. 2022 - May 2023*

- Conducted independent research on MRI reconstruction methods during a research internship, benchmarking supervised and self-supervised deep learning techniques.
- Collaborated with a team of researchers, contributed to the evaluation of deep learning networks for accelerated MRI reconstruction, and effectively communicated findings to team members and supervisors.

HOMLI (YC S22) — *Athens, Greece*

Data Science Intern — *Jun. 2022 - Aug. 2022*

- Successfully developed a model to calculate transfer time between properties and major destinations, resulting in a 30% reduction in Google service expenses.
- Implemented property price estimation models, created data preprocessing methods for Portuguese listing sites, and created automated analytics reports for data-driven decision-making in Attica's neighborhoods.

Publications

Reproduction and extension of "Robust Fair Clustering: A Novel Fairness Attack and Defense Framework"

[code] [paper]

- Reproduced and extended the ICLR'23 paper by Chhabra et al., refining the codebase, introducing additional metrics and datasets, and exploring novel attack strategies.
- Conducted comprehensive evaluations, including targeted attacks and an ablation study, to validate original claims and broaden the research landscape.

Notable Projects

Reproduction and extension of "Efficient Equivariant Transfer Learning from Pretrained Models"

[code] [blogpost]

- Reproduced and extended the original paper by Basu et al. (2023), verifying results across missing experiments and expanding the discussion on the weight patterns learned by λ-equitune.
- Proposed and implemented equiattention, an extension utilizing an Attention layer to potentially improve performance over λ-equitune.

Deep learning and Explainable Artificial Intelligence techniques for Alzheimer's disease detection on MRI

[paper]

- Developed an Alzheimer's disease detection framework using deep learning, transfer learning, and explainable AI on 2180 MRI images from the ADNI dataset.
- Implemented a comprehensive preprocessing pipeline and improved model explainability with Grad-CAM.

Real-Time Twitter Analysis

[code]

- Developed a Java-based web application for real-time Twitter data analysis using the Hadoop ecosystem.
- Gathered, processed, and analyzed Twitter data, utilizing Kafka, Avro, HDFS, and the Apache Mahoot K-Means Algorithm for clustering.

Education

University of Amsterdam — *Amsterdam, Netherlands*

MSc in Artificial Intelligence — *Sep. 2023 - Present*

- Full-time student at a top European university, specializing in Machine Learning, AI, Computer Vision, and Large Language Models (LLMs).

University of West Attica — *Athens, Greece*

BSc & MSc in Informatics & Computer Engineering (5-year degree; 300 ECTS) — *Sep. 2015 - Sep. 2022*

- Grade: 8.17/10 "Very Good"; (Graduated 3rd of my class)
- Thesis Project: "Deep learning and Explainable Artificial Intelligence techniques for Alzheimer's disease detection on MRI"; Grade: 10/10